

Practical No. 2

Aim: To study and implement DDL commands in RDBMS.

Software Requirement: MySQL, Oracle 10g

Theory:

DDL - Data Definition Language

DDL consists of the SQL commands that can be used for defining, altering and deleting database structures such as tables, indexes and schemas. It deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database.

1. CREATE Command:

CREATE command is used to create database or its objects (table, index, function, views, store procedure, and triggers)

Syntax: **CREATE TABLE** table_name (column1 data_type, column2 data_type, ...);

```
mysql> create table admin(ID int, Name varchar(50));
Query OK, 0 rows affected (7.27 sec)

mysql> desc admin;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int           | YES  |     | NULL    |       |
| Name  | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (5.69 sec)
```

2. ALTER Command:

ALTER command is used to alter the structure of the database.

Syntax: **ALTER TABLE** table_name ADD COLUMN column_name data_type;

```
mysql> alter table admin add email varchar(50);
Query OK, 0 rows affected (3.60 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc admin;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int           | YES  |     | NULL    |       |
| Name  | varchar(50)   | YES  |     | NULL    |       |
| email | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

3. RENAME Command:

RENAME command is used to rename an object existing in the database.

Syntax: **RENAME** TABLE old_table_name TO new_table_name;

```
mysql> rename table admin to administrator;
Query OK, 0 rows affected (1.19 sec)

mysql> desc administrator;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int           | YES  |     | NULL    |       |
| Name  | varchar(50)   | YES  |     | NULL    |       |
| email | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

4. TRUNCATE Command:

TRUNCATE command is used to remove all records from a table, including all spaces allocated for the records.

Syntax: **TRUNCATE** TABLE table_name;

```
mysql> truncate table administrator;
Query OK, 0 rows affected (1.58 sec)

mysql> desc administrator;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID    | int           | YES  |     | NULL    |       |
| Name  | varchar(50)   | YES  |     | NULL    |       |
| email | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

5. DROP Command:

DROP Command is used to delete objects from the database.

Syntax: **DROP** TABLE table_name;

```
mysql> drop table administrator;
Query OK, 0 rows affected (0.67 sec)

mysql> desc administrator;
ERROR 1146 (42S02): Table 'uni.administrator' doesn't exist
```

Conclusion: Hence, we have successfully studied and implemented DDL commands in RDBMS.